Basic Probability

Def. $X_n \xrightarrow{\mathscr{D}} X$ if their cdfs converge pointwise to the cdf of X. $X_n \xrightarrow{\text{p}} X$ if $\forall \epsilon > 0$: $\lim \mathbb{P}(|X_n - X| \geq \epsilon) = 0$.

Prop. Let F be the CDF of X. If F is continuous and strictly increasing, then $Y := F(X) \sim \text{Unif}[0,1]$. If $U \sim \text{Unif}[0,1]$, then $F^{-1}(U) \sim F$.

Prop. $Y := h(X)$. If h is injective and differentiable, then

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{\mathrm{d}h^{-1}(y)}{\mathrm{d}y} \right|.$$

Properties of $\mathbb{E}$, Var, and Cov:

- If $\int_{\mathbb{R}} |x| f(x)\,\mathrm{d}x < \infty$ then $\mathbb{E}(X) = \int_{\mathbb{R}} x f(x)\,\mathrm{d}x$.
- $\mathbb{E}$ commutes with continuous linear operators.
- If X is stochastically dominated by Y then $\mathbb{E}(X) \leq \mathbb{E}(Y)$.
- $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$.
- $\text{Var}(a + bX) = b^2 \text{Var}(X)$.
- $\text{Cov}(a + bX, c + dY) = bd\,\text{Cov}(X, Y)$.
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$.

If X and Y are independent:

- $\mathbb{E}(XY) = \mathbb{E}(X)\,\mathbb{E}(Y)$, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.
- $\mathbb{E}(X|Y) = \mathbb{E}(X)$ wp 1.
- X and any function of Y are independent.
- $\text{Cov}(X, Y) = 0$. The converse is not true: if $Z \sim N(0,1)$, and S and T are random signs (1 or -1), then $\text{Cov}(SZ, TZ) = 0$.

Prop. (i) Law of total expectation/Tower law: $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$.

(ii) We have the following wp 1:

$$\mathbb{E}[g(X)h(Y)|Y] = h(Y)\,\mathbb{E}(g(X)|Y), \mathbb{E}[X|T(Y)] = \mathbb{E}[\mathbb{E}[X|T(Y)]|Y].$$

(iii) Law of total variations: we have $\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}[\mathbb{E}(Y|X)]$, where $\text{Var}(Y|X) := \mathbb{E}(Y^2|X) - (\mathbb{E}(Y|X))^2$.

Prop. If $X \geq 0$ and there exists an at most countable subset $S = \{x_1, x_2, \dots\}$ of isolated points such that F_X is C^1 on $[0, \infty) \setminus S$, then

$$\mathbb{E}(X) = \sum_{x \in S} x\mathbb{P}(X = x) + \int_0^\infty x F_X'(x)\,\mathrm{d}x.$$

Joint Distribution

Def (Jacobian). $Jg(y) := \det\left[\partial g_i / \partial y_j\right]$.

Def (Conditional density). $f_{X|Y}(x|y) := f_{X,Y}(x, y)/f_Y(y)$.

Prop. If $X : \Omega \to H \subset \mathbb{R}^n$ and $h : H \to G \subset \mathbb{R}^n$, where H and G are open, are such that h is injective and differentiable and $h^{-1} : G \to H$ is differentiable. Then $Y := h(X)$ has density

$$f_Y(y) = \begin{cases} f_X(h^{-1}(y)) \cdot \left|Jh^{-1}(y)\right|, & y \in G \\ 0, & y \notin G. \end{cases}$$

Probability Inequalities

Thm (Ineqs). • Markov $X \geq 0$, $c > 0$, then $\mathbb{P}(X \geq c) \leq \frac{\mathbb{E}(X)}{c}$. Equality when $\mathbb{P}(X = 0 \text{ or } X = c) = 1$.

- Chebychev If $c > 0$, then $\forall \mu$: $\mathbb{P}(|X - \mu| \geq c) \leq \frac{\mathbb{E}[(X-\mu)^2]}{c^2}$.
- Chernoff If $c \in \mathbb{R}$ and $t > 0$, then

$$\mathbb{P}(X \geq c) \leq e^{-tc}\,\mathbb{E}(e^{tX}), \quad \mathbb{P}(X \leq c) \leq e^{tc}\,\mathbb{E}(e^{-tX}).$$

- Jensen $f : \mathbb{R} \to \mathbb{R}$ convex, $\mathbb{E}\,X < \infty$, then $f(\mathbb{E}\,X) \leq \mathbb{E}\,f(X)$.

Applications of Jensen:

- If f concave, then $f(\mathbb{E}\,X) \geq \mathbb{E}\,f(X)$.
- $(\mathbb{E}\,X)^2 \leq \mathbb{E}\,X^2$ and $\log \mathbb{E}\,X \geq \mathbb{E} \log X$.
- If $x_1, x_2, \dots, x_n > 0$, $p_i \geq 0$, and $\sum p_i = 1$, then $\prod x_i^{p_i} \leq \sum p_i x_i$. When $p_i = 1/n$, this is AM-GM.
- **Hölder**. $p, q > 0$, $\frac{1}{p} + \frac{1}{q} = 1$, then $\mathbb{E}\,XY \leq (\mathbb{E}\,X^p)^{\frac{1}{p}}\,(\mathbb{E}\,Y^q)^{\frac{1}{q}}$.
- **Cauchy**. $\mathbb{E}|XY| \leq \sqrt{\mathbb{E}\,X^2}\sqrt{\mathbb{E}\,Y^2}$.

Asymptotic Behavior

Let X_i iid with finite $\mathbb{E}\,X_i = \mu < \infty$ and $\text{Var}\,X_i = \sigma^2$. Then:

Thm (WLLN). $\overline{X}_n \xrightarrow{\text{p}} \mu$.

Thm (CLT). $\sqrt{n}(\overline{X}_n - \mu) \xrightarrow{\mathscr{D}} \mathcal{N}(0, \sigma^2)$.

Thm (Large Deviations). $c > \mu$. Then $\lim_{n\to\infty} \frac{1}{n} \log \mathbb{P}(\overline{X}_n > c) = -\sup_t [tc - \kappa(t)]$, where $\kappa(t) = \log \mathbb{E}(e^{tX})$.

Point Estimates

Bias: $\mathbb{E}(\hat{\lambda}) - \lambda$. Variance: $\text{Var}[\hat{\lambda}]$. $\text{MSE}[\hat{\lambda}] = \mathbb{E}[(\hat{\lambda} - \lambda)^2] = \text{Bias}^2 + \text{Var}$.

Def (Consistency). $\hat{\theta}_n \xrightarrow{\text{p}} \theta$.

Example. Modeling lifetime $T : \Omega \to [0, \infty)$.

- The **survival** function: $S : [0, \infty) \ni x \mapsto \mathbb{P}(T > x) = 1 - F_Y(x) \in [0, 1]$.
- The **failure rate** function: $h(x) := f(x)/S(x)$. Note that $\mathbb{P}(T \leq x + \Delta x | T > x) \approx \Delta x \cdot h(x)$.

Given h we can recover f: $f(x) = h(x) \exp\left(-\int_0^x h\right)$. If $h(x) = \lambda$ is constant, then $T \sim \text{Exp}(\lambda)$.

Method of Moments.

Thm (Continuous Mapping). g cont. Then $X_n \xrightarrow{\text{p}} X \implies g(X_n) \xrightarrow{\text{p}} g(X)$ and $X_n \xrightarrow{\mathscr{D}} X \implies g(X_n) \xrightarrow{\mathscr{D}} g(X)$.

Lem (Slutsky). If $X_n \xrightarrow{\mathscr{D}} X$ and $Y_n \xrightarrow{\text{p}} c$, where c is a constant, then $X_n + Y_n \xrightarrow{\mathscr{D}} X + c$ and $X_n Y_n \xrightarrow{\mathscr{D}} cX$.

Thm (Delta Method). If $\sqrt{n}(X_n - \theta) \xrightarrow{\mathscr{D}} \mathcal{N}(0, \sigma^2)$ and $g \in C^1$, then $\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{\mathscr{D}} \mathcal{N}(0, \sigma^2[g'(\theta)]^2)$.

Properties of MoM estimators:

- Not unique.
- Consistency by WLLN and the continuous mapping theorem.
- Asymptotic normality by CLT and Delta Method.

Likelihood Theory.

Def (Score Fn). $\dot{\ell}_n(\theta) := \frac{\partial}{\partial\theta}\ell_n(\theta) = \sum_{i=1}^n \frac{\frac{\partial f}{\partial\theta}(x_i, \theta)}{f(x_i, \theta)} = \sum_{i=1}^n \frac{f'(x_i, \theta)}{f(x_i, \theta)}$.

Prop. $\mathbb{E}_{\theta_0}\left[L_n(\theta)/L_n(\theta_0)\right] = 1$. $\mathbb{E}\left[\exp\left(\ell_n(\theta) - \ell_n(\theta_0)\right)\right] = 1$.

Prop. $\mathbb{E}_\theta\left[\dot{\ell}_n(\theta)\right] = 0$. $-\mathbb{E}_\theta\left[\ddot{\ell}_n(\theta)\right] = \mathbb{E}[\dot{\ell}_n(\theta)^2]$.

Def (Fisher Info). $I(\theta) := \mathbb{E}_\theta[\dot{\ell}(\theta)^2] = \mathbb{E}_\theta[-\ddot{\ell}(\theta)]$. That is, $I(\theta) := \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f(X, \theta)\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log f(X, \theta)\right]$. Expectation taken wrt $X \sim f(x, \theta)$.

Thm (Cramér–Rao Inequality). Let $T(X_n)$ be an unbiased estimator for $g(\theta)$. Then, $\text{Var}[T(X_n)] \geq [g'(\theta)]^2/nI(\theta)$.

Equality is attained if and only if $\text{Corr}(\hat{\theta}(X), \dot{\ell}(X)) = 1$, i.e., if $\hat{\theta}(X)$ and $\dot{\ell}(X)$ are linearly related rvs. That is, $\dot{\ell}(\theta) = \alpha(\theta)\hat{\theta}(X) + \beta(\theta)$ for functions α and β that do not depend on X.

Prop. There exists an unbiased estimator $\hat{\theta}(X)$ of θ whose variance attains the Cramér–Rao lower bound if and only if the score can be expressed in the form $\dot{\ell}(\theta) = I(\theta)\left\{\hat{\theta}(X) - \theta\right\}$ or, equivalently, if and only if the function $\frac{\dot{\ell}(\theta)}{I(\theta)} + \theta$ does not depend on θ.

Thm (Fisher). $X_1, \dots, X_n \overset{\text{iid}}{\sim} f(x|\theta_0)$. As $n \to \infty$, we have

$$\sqrt{nI(\theta_0)} \cdot (\hat{\theta} - \theta_0) \xrightarrow{\mathscr{D}} \mathcal{N}(0, 1) \quad \text{and} \quad \sqrt{nI(\hat{\theta})} \cdot (\hat{\theta} - \theta_0) \xrightarrow{\mathscr{D}} \mathcal{N}(0, 1)$$

Think also $\hat{\theta} \approx \mathcal{N}\left(\theta_0, 1/nI(\theta_0)\right)$.

Prop. Assumptions: densities are smooth, have common support, and are distinct for different parameters. Properties of MLE under the above assumptions: consistent, asymptotically normal, known and

optimal asymptotic variance (**efficiency**), invariance (if $\hat{\theta}_n$ is an MLE of θ, then $\hat{\tau}_n := g(\hat{\theta}_n)$ is an MLE of $g(\theta)$).

Confidence Intervals

If $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, n small: Recall exact distribution results.
CI for μ: $\left[\overline{X} + \frac{t_{L,n-1}S}{\sqrt{n}}, \overline{X} + \frac{t_{R,n-1}S}{\sqrt{n}}\right]$. CI for σ^2: $\left[\frac{(n-1)S^2}{\chi^2_{R,n-1}}, \frac{(n-1)S^2}{\chi^2_{L,n-1}}\right]$.
If $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, n large: use CLT.
CI for μ: $\left[\overline{X} - \frac{z_R\sigma}{\sqrt{n}}, \overline{X} - \frac{z_L\sigma}{\sqrt{n}}\right]$ (use S to approximate σ if needed).
If $\hat{\theta}$ is MLE: use Fisher.
CI for θ: $\left[\hat{\theta} - \frac{z_R}{\sqrt{nI(\hat{\theta})}}, \hat{\theta} - \frac{z_L}{\sqrt{nI(\hat{\theta})}}\right]$.

The Bootstrap

Def (Empirical Dsitribution Function)**.** Let $X_i \overset{\text{iid}}{\sim} F$. The **edf**, $\hat{F}_n$, is the cdf that puts probability $1/n$ at each X_i. $\hat{F}_n(x) := \frac{1}{n}\sum \mathbb{1}_{\{X_i \le x\}}$.
Properties of the edf:

- $\mathbb{E}[\hat{F}(x)] = F(x)$.
- $\mathrm{Var}(\hat{F}(x)) = F(x)(1 - F(x))/n$.
- By WLLN, $\hat{F}(x) \xrightarrow{\text{p}} F(x)$.
- $\hat{F}_n(x) \to F(x)$ uniformly. $\sup_x \left|\hat{F}_n(x) - F(x)\right| \to 0$.

Hypothesis Testing

Power: $\pi : \Omega \to [0, 1]$, $\pi(\theta) := \mathbb{P}_\theta(x \in S_1)$. The **size** of a test: $\sup_{\theta\in\Omega_0} \pi(\theta)$. **Level α test**: size $\le \alpha$.
Prop. If under H_0, $T \sim F$, then $p = \mathbb{P}(T \ge T_{\text{obs}})$. Moreover, $F(p) \sim \text{Uniform}[0, 1]$.

Likelihood Ratio Test

Thm (Neyman-Pearson Lemma)**.** With LR and S_1 as above, if the type I and type II errors are α and β, then any other test with α type I error has a type II error larger than β.
Def. For $H_0 : \theta \in \Omega_0$ and $H_A : \theta \in \Omega_1$, we can define the **likelihood ratio** as follows: $\Lambda(X) := \frac{\sup_{\theta\in\Omega_0} L(\theta|X)}{\sup_{\theta\in\Omega} L(\theta|X)}$.
Thm. If $\Omega \subset \mathbb{R}^p$ open and Ω_0 is obtained by fixing k coordinates of θ and if assumptions of MLE hold, then under H_0 we have $-2\log\Lambda(X) \xrightarrow{\mathcal{D}} \chi^2_k$.

Multiple Testing

m independent tests. To obtain $\mathbb{P}[V > 0] < \alpha$, where V is the number of false discoveries (rejection under H_0): Sidak correction: reject when $p < \gamma := 1 - (1 - \alpha)^{1/m}$. Bonferroni: $\gamma := \alpha/m$.
Algo (Benjamini Hochberg procedure, 1985)**.**

- Sort all p-values in ascending order. $p_{(1)} \le \cdots \le p_{(m)}$.
- Find the largest j such that $p_{(j)} \le \frac{\alpha j}{m}$.
- Reject the tests with the j smallest p-values.

Bayesian Statistics

Prior	Likelihood	Posterior
$\text{Gam}(\alpha, \beta)$	$\text{Poi}(\lambda)$	$\text{Gam}(\alpha + n\bar{X}, \beta + n)$
$\text{Gam}(\alpha, \beta)$	$\text{Exp}(\lambda)$	$\text{Gam}(\alpha + n, \beta + n\bar{X})$
$\text{Beta}(\alpha, \beta)$	$\text{Bernoulli}(\lambda)$ or $\text{Bin}(n, \lambda)$	$\text{Beta}(\alpha + n, n - n\bar{X} + \beta)$
$\mathcal{N}(\mu_0, \nu^2)$	$\mathcal{N}(\mu, \sigma^2)$ (σ known)	$\mathcal{N}\left(\frac{\frac{\mu_0}{\nu^2} + \frac{n\bar{X}}{\sigma^2}}{\frac{1}{\nu^2} + \frac{n}{\sigma^2}}, \frac{1}{\frac{1}{\nu^2} + \frac{n}{\sigma^2}}\right)$

Def. The **Jeffreys** prior is defined as $\pi_J(\theta) \propto \sqrt{I(\theta)}$.
Invariant under reparameterization.

Statistical Decision Theory

Frequentist risk $R(\theta, \hat{\theta}) := \mathbb{E}_\theta\, L(\theta, \hat{\theta}) = \int L(\theta, \hat{\theta}) f(X|\theta)\, \mathrm{d}X$. If L is squared loss, then $R(\theta, \hat{\theta}) = \text{MSE}(\hat{\theta})$.
Def. Maximum risk: $\overline{R}(\hat{\theta}) = \max_\theta R(\theta, \hat{\theta})$. A **minimax estimator** is an estimator $\hat{\theta}$ such that $\overline{R}(\hat{\theta}) = \inf_{\tilde{\theta}} \overline{R}(\tilde{\theta})$.
Def. With π prior, the **Bayes risk** is $r(\hat{\theta}) = \int R(\theta, \hat{\theta})\pi(\theta)\, \mathrm{d}\theta$. A **Bayes estimator** (associated with L, π) is an estimator $\hat{\theta}$ such that $r(\hat{\theta}) = \min_{\tilde{\theta}} r(\tilde{\theta})$.
Def. The **posterior risk** is risk calculated using the posterior distribution: $r(\hat{\theta}|X) = \int L(\theta, \hat{\theta}) f(\theta|X)\, \mathrm{d}\theta$.
Thm. Let $\hat{\theta} = \hat{\theta}(X)$ be the value that minimizes the posterior risk $r(\hat{\theta}|X)$. Then $\hat{\theta}$ is the Bayes estimator.

L	squared error	absolute error	zero-one
Bayes estimator	posterior mean	$\sim$ median	$\sim$ mode

Distributions and Properties

$X \sim$	$\text{supp}\,\mathbb{P}(X = \cdot)$	$\mathbb{P}(X = x)$	$\mathbb{E}$	Var
$\text{Bin}(n, p)$	$\{0, 1, \ldots, n\}$	$\binom{n}{x} p^x (1 - p)^{n-x}$	np	$np(1 - p)$
$\text{Geom}(p)$	$\mathbb{N}$	$(1 - p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$\text{Poi}(\lambda)$	$\mathbb{N} \cup \{0\}$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ

$X \sim$	supp f	f	$\mathbb{E}$	Var
$\text{Unif}(a, b)$	$[a, b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\mathcal{N}(\mu, \sigma^2)$	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
$\text{Exp}(\lambda), \text{Gam}(1, \lambda)$	$[0, \infty)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\text{Gam}(\alpha, \beta)$	$(0, \infty)$	$\frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
$\text{Beta}(\alpha, \beta)$	$(0, 1)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Properties of $\mathcal{N}$: Let $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Let $S^2 := \frac{1}{n-1}\sum(X_i - \overline{X})^2$.

- $\mathbb{E}[S^2] = \sigma^2$.
- $\overline{X}$ and S^2 are independent.
- $\sqrt{n} \cdot \frac{\overline{X}-\mu}{S} \sim t_{n-1}$ and $(n - 1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$.
- $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.
- If X and $Y \sim \mathcal{N}(\mu_X, \gamma_Y^2)$ are independent, then $X + Y \sim \mathcal{N}(\mu + \mu_Y, \sigma^2 + \sigma_Y^2)$.

Properties of Exp:

(i) "Memoryless": $\mathbb{P}(T \le x + y|T > x) = \mathbb{P}(T \le y)$.
(ii) $\text{Exp}(\lambda) = \text{Gam}(1, \lambda)$.

Properties of Gam:

(i) If $X_i \overset{\text{iid}}{\sim} \text{Gam}(\alpha_i, \beta)$, then $\sum X_i \sim \text{Gam}\left(\sum \alpha_i, \beta\right)$.
(ii) If $X \sim \text{Gam}(\alpha, \beta)$ and $\alpha > 1$, then $\mathbb{E}\left[1/X\right] = \frac{\beta}{\alpha-1}$.
(iii) If $X \sim \text{Gam}(\alpha, \beta)$, then $\beta X \sim \text{Gam}\left(\alpha, 1\right)$.

Miscellaneous

Gamma: $\Gamma(\lambda) := \int_0^\infty e^{-x} x^{\lambda-1}\, \mathrm{d}x$.
Properties: $\Gamma(1) = 1; \Gamma(1/2) = \sqrt{\pi}; \Gamma(x+1) = x\Gamma(x); \Gamma(n) = (n-1)!$, $\forall n \in \mathbb{N}$.
MGF: $M_X : \mathbb{R} \ni t \mapsto \mathbb{E}[e^{tX}] \in [0, \infty]$. Properties:

- $\mathbb{E}[X^n] = M_X^{(n)}(0)$ when Fubini grants. $\mathbb{E}[e^{tX}] = \sum_{n=0}^\infty \frac{t^n\, \mathbb{E}(X^n)}{n!}$.
- $M_{cX}(t) = M_X(ct)$.
- If X, Y independent, then $M_{X+Y}(t) = M_X(t) + M_Y(t)$.
- If X_i iid, then $M_{\sum X_i} = \prod M_{X_i}$.
- $X_n \xrightarrow{\mathcal{D}} X \iff M_{X_n} \to M_X$ in a neighborhood of 0.